

CSP502 Software Lab-I – M.Tech. CSE W26

Lab Exercise 4B: Data Structures as Design Decisions — Building and Evaluating a Real-World Software Component

In **Lab Exercise 4A**, you created and validated synthetic log files having different sizes and deliberately introduced workload characteristics.

Now, you have to use **your own generated and validated log files** to develop a **Linux Log/Event Analyzer**.

Here is a realistic problem for which there is no single obvious data structure. You must first make a design decision, implement it, test it, and then defend it using your workload and observed results.

Problem: Linux Log/Event Analyzer

Develop a program capable of answering queries such as:

1. Which users generated the most events?
2. How many unique users/IP addresses occurred?
3. Find all events belonging to a particular user.
4. Find events occurring within a given time interval.
5. Identify repeated failed logins.
6. Find the top k most frequently accessed files.
7. Identify users who were active during a specified interval.

The important part, however, comes **before coding**.

Phase 1 — Understand Your Workload and Make Data-Structure Decisions

First, go back to the validation results of your **Lab Exercise 4A**.

Identify the important characteristics of your own workload that may influence the design of the analyzer.

For example, consider:

- distribution of events among users;
- number of unique users/IP addresses;
- frequency of different event types;
- frequently accessed resources;
- distribution of events over time;
- duplicates and malformed records;
- any other characteristic that you deliberately introduced or subsequently observed.

Do not assume that all groups have the same workload characteristics.

For different operations, consider appropriate **sequential, associative, ordered, priority-based, or composite data structures**.

Prepare the following decision table **before implementation**:

Required Operation	Relevant Workload Characteristic from 4A	Candidate Structure(s)	Selected Structure	Expected Complexity	Reason
Lookup user					
Count occurrences					
Top- k events/resources					
Time-range search					
Any other important operation					

You should predict the expected complexity before implementation.

Important:

Submit/show this initial decision table to the instructor before proceeding to implementation.

You should not modify your original prediction after obtaining the experimental results. If your understanding changes later, record that separately as your **revised understanding**.

Phase 2 — Implementation

Implement the Log/Event Analyzer using your selected structures.

Consider the following constraint:

You cannot use one data structure indiscriminately for every operation.

Think about whether different queries warrant different representations or auxiliary structures. Your implementation should work with all three datasets generated in Practical 4A:

1,000 → 10,000 → 100,000 records

The program should also handle the duplicates and malformed records present in your dataset in a manner decided and documented by you.

Phase 3 — Compare Two Design Alternatives

Select **one important query/operation** from your analyzer for which at least two plausible data-structure strategies are possible.

Implement the operation using **two alternative approaches**.

Before running them, record:

1. Which approach do you expect to perform better?
2. Why?
3. Under what workload characteristics do you expect your prediction to hold?

Do not change this prediction after observing the results.

Phase 4 — Empirical Validation

Run your implementation with:

1,000 → 10,000 → 100,000 records

For the selected query/operation, compare the two alternative approaches.

Record at least:

- loading/construction time;
- query execution time;
- observed behaviour as the input size increases.

Memory usage may also be measured, where feasible.

Prepare a small comparison table:

Dataset Size	Approach 1	Approach 2	Expected Behaviour	Observed Behaviour
1,000				
10,000				
1,00,000				

After that, compare the observed behaviour against the complexity and performance that you predicted earlier.

And answer the following question:

Did the experimental observations support your original data-structure decision? If not, what did you learn from the difference?

Phase 5 — A Deliberate Design Disturbance

After about 1 hour 15 minutes, you will get a **new or changed requirement**.

Do not immediately start modifying the program.

First answer:

1. Which assumption underlying your original data-structure decision has been affected by the changed requirement?
2. Which part of your present data representation or auxiliary structures must change?
3. Which part can remain unchanged?
4. Does your existing design support the new requirement efficiently? Why or why not?

Record your answers **before modifying the implementation**.

After that, make the necessary design and implementation changes.

Validate whether the modified design satisfies the changed requirement.

Reflection Questions

R1. Which design decision had the greatest effect on your program's behaviour as the dataset increased?

R2. Did the theoretically preferable data structure always produce the practically preferable solution? What evidence supports your conclusion?

R3. Which assumption underlying your original data-structure decision was challenged by the changed requirement? How did that affect your design?

R4. If the dataset increased from 100,000 records to one billion records, which part of your present design would become unacceptable first? Why?

R5. Suppose an AI tool recommends a hash table because lookup is “O(1)”.

What information about the actual system, workload, operations, and constraints would you need before accepting that recommendation?

Submission

Each group should submit:

1. **Initial Data-Structure Decision Table** prepared before implementation.
2. Source code of the Log/Event Analyzer.
3. Source code/implementation of the two alternative approaches selected for comparison.
4. Experimental results for **1,000, 10,000 and 100,000 records**.
5. Record of the **original prediction and observed results**.
6. Response to the changed requirement, including:
 - what assumption changed;
 - what part of the design had to change;
 - what could remain unchanged;
 - evidence after modification.
7. Answers to the reflection questions.

Retain the datasets, programs, observations and performance results. They will be useful in the subsequent practical on **algorithm choice and performance analysis**.

Expected Learning

At the end of this practical, you should be able to see that:

A data structure is not inherently “best.” Its suitability depends upon the operations to be performed, characteristics of the workload, scale of the data, and requirements of the system.

More importantly, a data-structure decision should be **predicted, justified, implemented, measured and reconsidered using evidence**, rather than selected merely because of its theoretical complexity.
